

Calibrating the Force-Measurement System

Step-by-step guide — how and why the sensors are calibrated.

1. Why do we calibrate?

1.1. What the sensors actually measure

Each of the four boards contains a **three-axis load cell**: a small device that produces a tiny electrical signal when it is pushed or pulled. The boards are plugged into the computer over USB, and the backend reads all four about 100 times per second, delivering twelve numbers per reading to the dashboard (four boards $\times$ three directions, X , Y and Z). The catch is that these raw numbers are **voltage ratios**, not forces. They are extremely small (around 10^{-6}) and every individual sensor is slightly different. A reading of, say, 0.000012 from one board does not directly tell you how many newtons are pressing on it. We need a rule that converts the raw number into a force. **Finding that rule, for each sensor, is what calibration does.**

1.2. The conversion rule

For every sensor channel we assume a simple *straight-line* relationship between the raw reading and the true force:

$$\text{force (N)} = (\text{raw reading} - \text{offset}) \times \text{scale}. \tag{1}$$

Two unknowns appear here, and they behave *very differently* — which is why the system handles them in two separate ways:

- **scale** — how many newtons correspond to one unit of raw signal (the steepness, or slope, of the line). This is a *fixed physical property* of the sensor; it barely changes over time. You measure it **once**, with known weights, and it is **saved**. Finding the scale is what **calibration** means in this guide.
- **offset** — the raw number the sensor reports when *nothing* is loaded (its “zero point”). This one *drifts* — with temperature, as the rig settles, whenever the hardware is re-seated — so a value measured today is wrong tomorrow. Rather than save it, you re-set it **live, on demand**, with the Zero Sensors button (a “tare”). The offset is **never saved** as part of the calibration.

In short: the **scale** is stable, so it is calibrated and stored; the **zero** drifts, so it is tared fresh whenever you need it (Section 5).

1.3. Three things, kept separately

This split into a stable, saved *scale* and a live, unsaved *zero* is the one idea to hold on to. In full, the system keeps **three** things apart, each in its own place — and only the first two are ever saved:

What	What it is	Saved?
Active calibration	axis switches + per-channel <i>scales</i>	yes (durable)
Calibration profiles	a named library of saved calibrations	yes (durable)
Zero (tare)	the live per-channel zero point	no (live only)

The **active calibration** is the scale set currently in force — what you produce in the wizard. A **profile** is a named snapshot of an active calibration you can save and reload (Section 6), handy when you swap between walls or sensor sets. The **zero** is the live tare (Section 5): cached in your browser, but never sent to the backend and never part of a calibration or a profile.

1.4. Why known weights — and why three of them?

We create known forces by hanging known masses. A mass m produces a downward gravitational force

$$F = m g, \quad g = 9.80665 \text{ m/s}^2, \quad (2)$$

so 10 kg gives about 98 N and 20 kg about 196 N. By recording the raw sensor value at several known weights we can work out the two unknowns in Equation (1).

The standard procedure uses **three weights**: 0, 10 and 20 kg (you can change these values in the app). Why three?

- The 0 kg measurement gives the fit its **zero reference** — the reading with no load — so the slope is measured from the right baseline. (This reference is used only to work out the scale; it is *not* saved as the sensor's zero. Setting the live zero is a separate, on-demand step — Section 5.)
- A **single loaded weight** would already be enough to estimate the **scale** (slope) — geometrically, two points define a straight line.
- The **third weight** is what makes the result trustworthy. With three points we can *check* that the sensor really behaves like a straight line, and we can *average out* measurement noise instead of trusting a single reading. More points $\Rightarrow$ a more reliable slope.

In short: one point anchors the baseline, the others fix and verify the slope — and the slope is the only thing calibration keeps.

1.5. The method: a best-fit straight line

We do not just connect dots — we fit the **best straight line** through the measured points using the method of *least squares* (the line that sits as close as possible to all points at once). Because we already know the zero point from the 0 kg reading, the line is forced through the origin once the offset is subtracted. The slope of that line is the **scale**:

$$\text{scale} = \frac{\sum_k x_k F_k}{\sum_k x_k^2}, \quad x_k = (\text{raw}_k - \text{offset}), \quad (3)$$

where F_k is the known force at step k and x_k is the offset-corrected raw reading. This is exactly the formula the app uses internally; you never compute it by hand — the wizard does it the moment you finish capturing the weights.

This slope is not arbitrary: it is the one that makes the line fit the points as well as possible. We choose **scale** to minimise the total squared error between the known forces and the line's predictions,

$$S(\text{scale}) = \sum_k (F_k - \text{scale} \cdot x_k)^2. \quad (4)$$

Setting the derivative to zero, $\frac{dS}{d(\text{scale})} = -2 \sum_k x_k (F_k - \text{scale} x_k) = 0$, and solving for **scale** gives Equation (3). Since S is an upward-opening parabola in **scale** (its second derivative $2 \sum_k x_k^2$ is positive), this critical point is the unique minimum — the best-fit slope.

In one sentence: calibration applies known weights to each sensor, fits a straight line through the raw readings, and stores the line's **slope** (*scale*) so that every future reading can be turned into newtons. The line's zero point is *not* saved — you set it live with **Zero Sensors** (Section 5).

2. The wall is tilted — and why that helps

The climbing wall does not stand vertically; it **leans back** by an angle we call θ (theta). This tilt is not a nuisance — it is the key to a clever shortcut.

Each board measures force along two perpendicular directions that matter here:

- X — *along the wall surface* (the “in-plane” or up-slope direction),
- Z — *perpendicular to the wall surface* (the “normal” or out-of-wall direction),
- Y — sideways (left/right), handled separately.

When you hang a weight straight down on a wall that is tilted by θ , gravity no longer points purely along one axis. Instead the single vertical force $W = mg$ splits into two parts (Figure 1):

$$\underbrace{W \cos \theta}_{\text{along the wall (X)}} \quad \text{and} \quad \underbrace{W \sin \theta}_{\text{out of the wall (Z)}} . \quad (5)$$

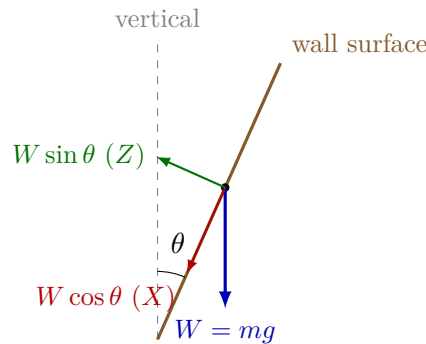


Figure 1. On a wall tilted by θ from vertical, one hanging weight W loads both the along-wall axis (X , by $W \cos \theta$) and the out-of-wall axis (Z , by $W \sin \theta$) at the same time.

This is the so-called **single-hang method**: because the tilt automatically spreads the load across both the X and Z axes, **one set of hangs calibrates two axes at once**. You only have to tell the app the wall’s tilt angle θ so it knows how the weight was divided. (The third axis, Y / sideways, is calibrated with a separate left/right pull.)

The angle θ must match the real wall. θ is used only to split a hang into its X and Z parts. If you type the wrong angle, the split is wrong and the calibration comes out wrong. Measure the wall’s lean-back before you start (a phone inclinometer app is fine). At $\theta \approx 0^\circ$ a hang puts no load on Z at all, so the wall *must* be tilted for this method to work.

2.1. The coordinate convention

Throughout the system the directions are fixed as:

Axis	Positive direction
$+X$	up (toward the top of the wall)
$+Y$	to the right
$+Z$	out of the wall (toward the climber)

A climber simply hanging from a hold therefore registers a *negative* X (they pull down-slope), and a pull away from the wall registers a *positive* Z .

3. Before you begin

You will need:

- the backend running and the dashboard open in a browser (see the *User Guide* for how to start them);

- the four sensor boards **plugged in** (via the USB hub) and the dashboard **connected** (click **Connect Device** in the toolbar) — calibration reads live data, so without a connection every capture will time out;
- **known weights**, e.g. 10 kg and 20 kg plates, plus the ability to hang/pull them cleanly on each board;
- the wall's **tilt angle** θ , measured in degrees from vertical.

You calibrate **four boards**, and for each board you calibrate the **hang** (X & Z) and the Y (**sideways**) direction. That is the full procedure, and it sets the **scales** only. Axes you skip keep rough factory-default values and may report inaccurate forces. Zeroing (the tare) is a separate one-click step done from the main toolbar and needs no weights — [Section 5](#).

4. Walkthrough: calibrating step by step

Open the wizard by clicking **Calibrate** in the toolbar's *sensor-calibration* tray (alongside **Zero Sensors** and **Profiles**). The wizard sets the **scales**, and walks you through five stages:

Overview → Set angle θ → Pick a board → Pick what to calibrate → Capture & save.

4.1. Stage 1 — Overview

The first screen (Figure 2) summarises the procedure and reminds you that hanging one weight calibrates both *X* and *Z*. If you see an amber “No device connected” banner, plug in and connect the device first. Click **Start Calibration** to continue.

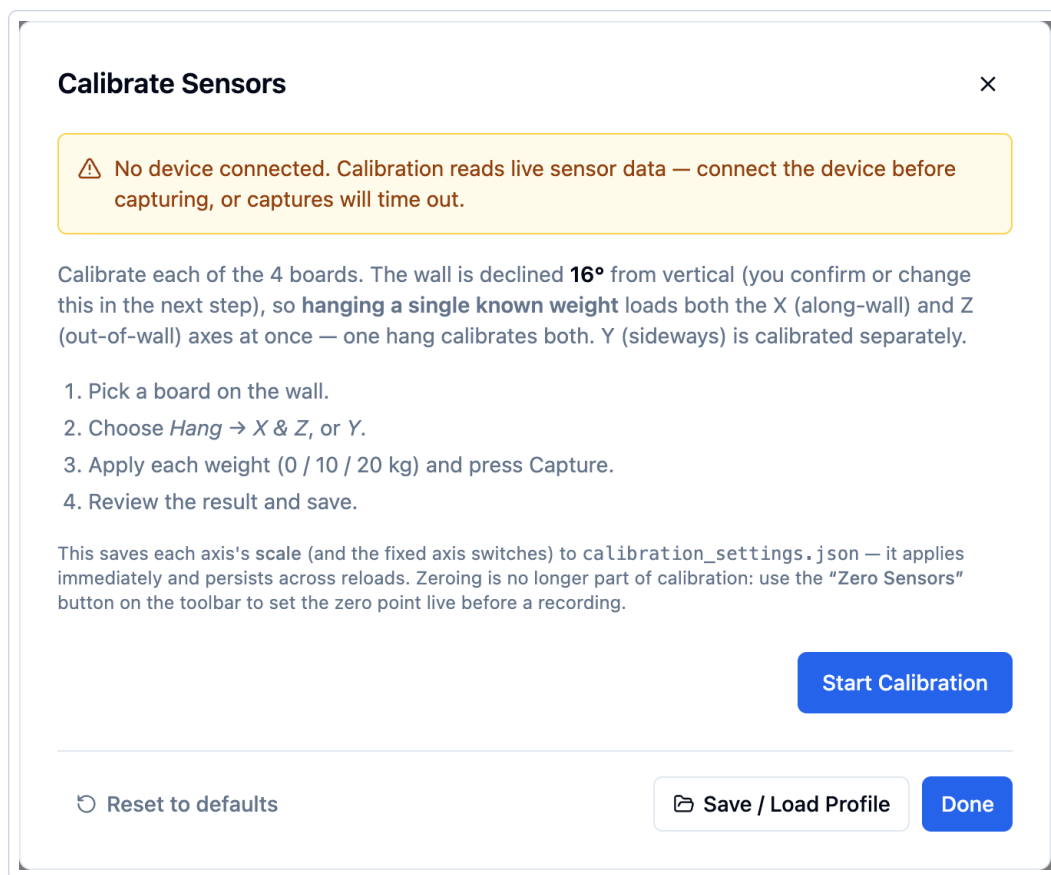


Figure 2. Stage 1 — the overview screen explains the procedure before you begin.

4.2. Stage 2 — Set the wall angle θ

Enter the wall's true lean-back from vertical (Figure 3). This is the *only* place the calibration angle can be set, on purpose: it keeps the saved *X/Z* calibration and the angle from ever drifting apart. The default is 16°. Type your measured value and click **Next** — pick a board. The dashboard's **View in world frame** dialog asks for a tilt as well, but that is a *separate, display-only* angle — pre-filled from this one and never written back to it.

←

Calibrate Sensors

×

⚠ No device connected. Calibration reads live sensor data — connect the device before capturing, or captures will time out.

Step 1 · Set the wall decline angle θ

Set this to your wall's true lean-back from vertical. It splits a single vertical hang into the X (along-wall) and Z (out-of-wall) axes, so the boards you calibrate next are computed for exactly this angle.

⚠

Wall decline angle θ

How far the wall leans back from vertical (–90° to 90°).

16

° from vertical

⚠ θ defines how X & Z are split from a hang. If you change it, recalibrate each board's Hang → X & Z so the directional forces stay correct. The overall force magnitude and Y are unaffected.

Back

Next — pick a board

⌛ Reset to defaults

📁 Save / Load Profile

Done

Figure 3. Stage 2 — enter the measured wall tilt θ . It tells the app how a hang divides between the X and Z axes.

4.3. Stage 3 — Pick a board

A 2×2 grid represents the wall as you face it: hands on top, feet on the bottom (Figure 4). Each board shows three little badges, X, Y and Z. A **green check** means that axis is already calibrated; a plain letter means it still uses factory defaults. Click the board you want to work on.

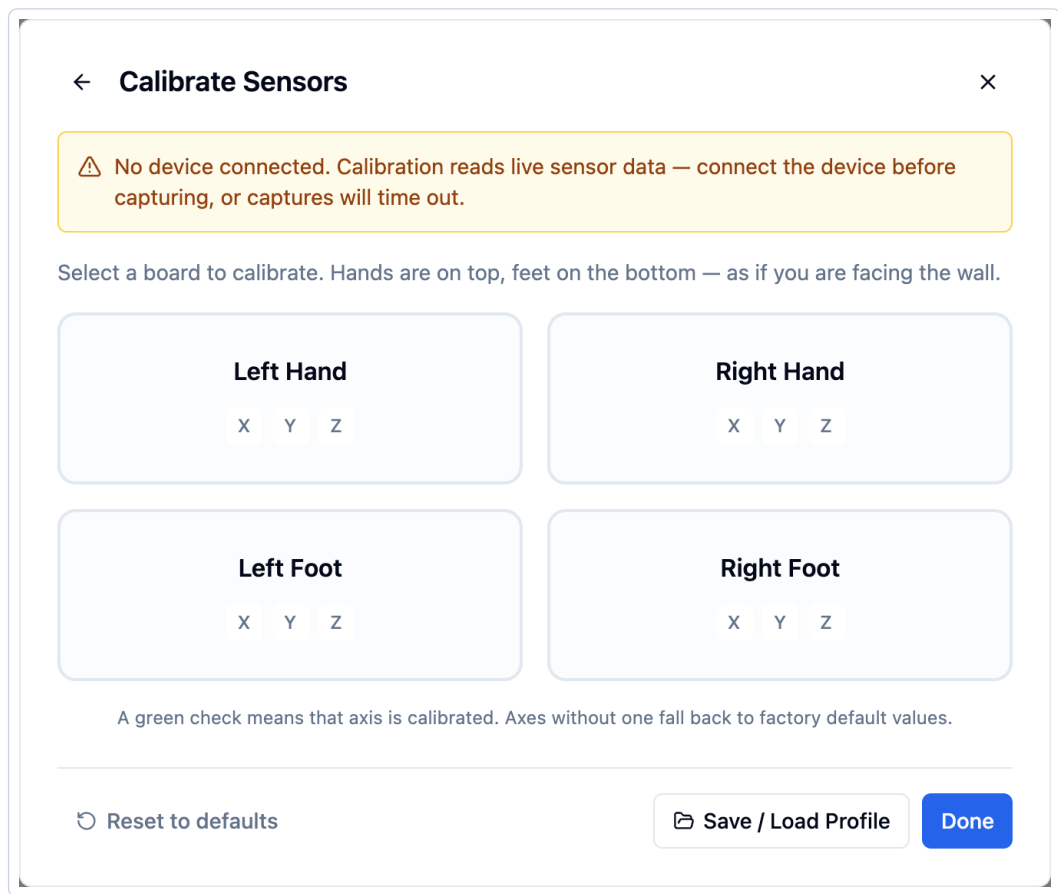


Figure 4. Stage 3 — choose a board. Green checks mark axes that are already calibrated.

4.4. Stage 4 — Pick what to calibrate

Now choose *what* to calibrate on this board (Figure 5):

- **Hang** → *X* & *Z* — the single-hang method. You hang weights straight down and the app calibrates both the along-wall (*X*) and out-of-wall (*Z*) axes together. (Disabled if $\theta \approx 0$.)
- *Y* — **sideways** — you pull the weight left or right to calibrate the sideways axis. The app knows that left-side boards are pulled left and flips the sign automatically.

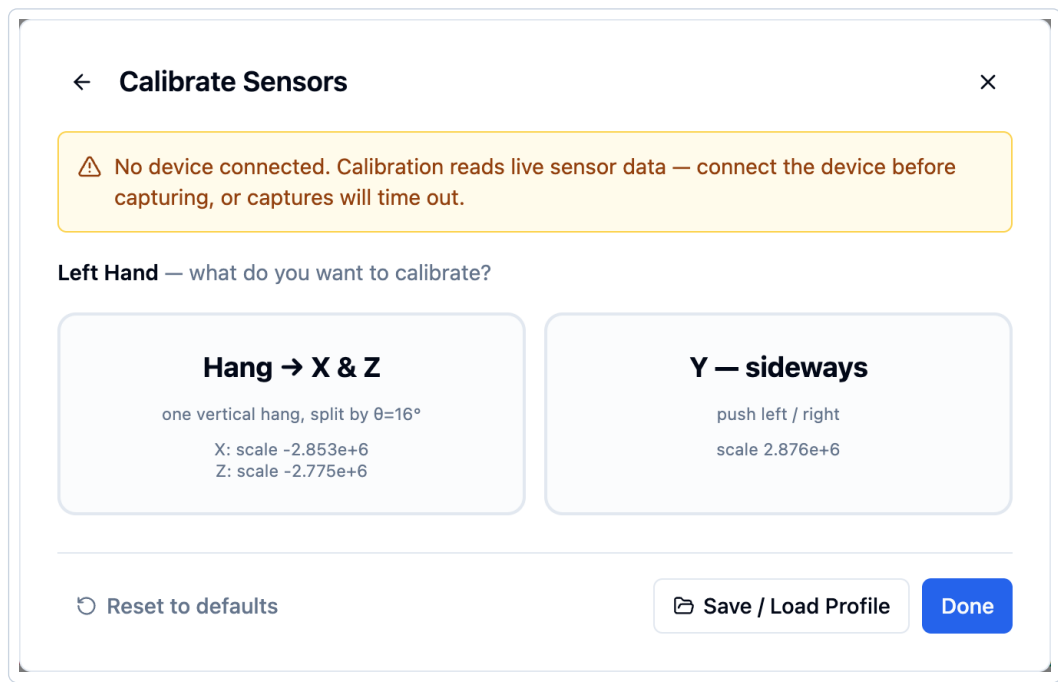


Figure 5. Stage 4 — calibrate the hang (X & Z) or the sideways (Y) direction for the selected board.

4.5. Stage 5 — Capture the weights and save

This is where the measurements happen (Figure 6). You will see three rows for 0, 10 and 20 kg (these numbers are editable). For each row:

1. Apply the stated weight. For a **hang**, let the weight hang *straight down* and stay still. For Y , pull it cleanly to the side.
2. Wait for it to stop moving, then click **Capture**. The app averages about 1.5 seconds of live data (~150 readings) to get a clean value.
3. Repeat for all three weights. The 0 kg step (nothing applied) gives the fit its zero reference — it is used to work out the scale, not saved as the sensor’s zero.

A live “**Steady / Swinging**” indicator tells you whether the weight is still moving. If you capture while it swings, the row is flagged “swung” — you are never blocked, but a steadier capture is more accurate, so re-capture once it settles.

Once all three weights are captured, the app instantly computes and displays the resulting **scale** (for a hang, one for X and one for Z) — no offset is shown or stored. Review the green result panel, then click **Save**. The calibration takes effect immediately and is stored permanently (see Section 7). You are returned to the board’s mode picker so you can calibrate its other axis, or go back and pick another board.

← **Calibrate Sensors** ×

⚠ No device connected. Calibration reads live sensor data — connect the device before capturing, or captures will time out.

Calibrating **Left Hand · X & Z (hang)** Live raw: **X 0.000e+0 · Z 0.000e+0**

Hang the known weight straight down. On the 16°-declined wall this loads the along-wall axis by $W \cdot \cos 16^\circ$ ($\rightarrow X$) and the out-of-wall axis by $W \cdot \sin 16^\circ$ ($\rightarrow Z$), so this one hang calibrates both. A hanging climber reads $-X$ (vertical); a pull out of the wall reads $+Z$.

Step 0 kg not captured **Capture**

Step 1 kg not captured **Capture**

Step 2 kg not captured **Capture**

Back **✓ Save X & Z**

Reset to defaults **Save / Load Profile** **Done**

Figure 6. Stage 5 — apply each known weight and click Capture. The app fits the line and shows the computed scale before you save.

4.6. Finishing

Click **Done** when every board and axis is calibrated. If any axis is still on factory defaults, the app warns you and lists exactly which ones, so nothing is forgotten. There is also an **Exit & discard** option that rolls back every change you made during the session (including θ), and a **Reset to defaults** button at the bottom that clears *all* calibration.

Two things worth knowing about the wizard. The **green “calibrated” checks** start empty every time you open it and are filled only by saves you make *in that session* — they track this sitting, not your whole history. And the footer’s **Save / Load Profile** button opens the profile manager (Section 6) without leaving the wizard.

5. Zeroing the sensors (tare)

Calibration fixes the *scale*; **zeroing** fixes the *zero point*. Because the zero drifts, you re-set it whenever you start measuring — it takes one click, needs no weights, and never touches your calibration.

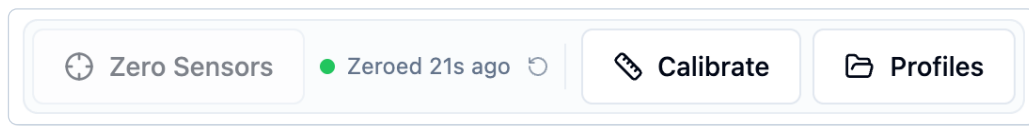


Figure 7. The sensor-calibration tray: Zero Sensors with its live status chip (Zeroed *X*s ago and a freshness dot), next to Calibrate and Profiles.

How to zero.

1. Make sure the holds are **unloaded** (nothing hanging, nobody on the wall) and the device is connected and streaming.
2. Click **Zero Sensors** in the toolbar's sensor-calibration tray. The app averages about 1.5 seconds of the most recent readings (~150 samples) and installs that as the new zero *instantly* — a recording in progress never pauses.

Reading the status. Next to the button a small chip reports the state: Not zeroed yet before you have zeroed, then Zeroed *X*s ago with a coloured dot — **green** under a minute, **amber** under five, red beyond — so you can tell at a glance whether the zero is fresh. The small  beside it clears the zero and returns to raw, un-tared readings.

The zero is **cached in your browser**, so it survives a page reload and you do not have to re-zero after refreshing. But it is **never sent to the backend** and is **never part of a calibration or profile**: it is the one value the system deliberately keeps live rather than saved.

Zeroing needs a live device. The button stays disabled until about 20 samples have streamed in, and **demo mode does not feed it** (demo data bypasses the sensor pipeline, the same limitation as the wizard's live readout). Connect a real device to zero.

When to zero. At the start of each session, and any time the unloaded reading looks off or the staleness dot turns red. You do *not* need to recalibrate in order to re-zero — keeping the two apart is exactly the point.

6. Calibration profiles

A **profile** is a named, saved copy of a full calibration (all the axis switches and scales — never the zero). Profiles let you keep more than one calibration on hand: one per wall, one per set of sensors, or a known-good backup before you experiment. They live on the backend and survive restarts.

Open the manager from the **Profiles** button in the toolbar, or from **Save / Load Profile** in the calibration wizard's footer — both open the same **Calibration Profiles** panel.

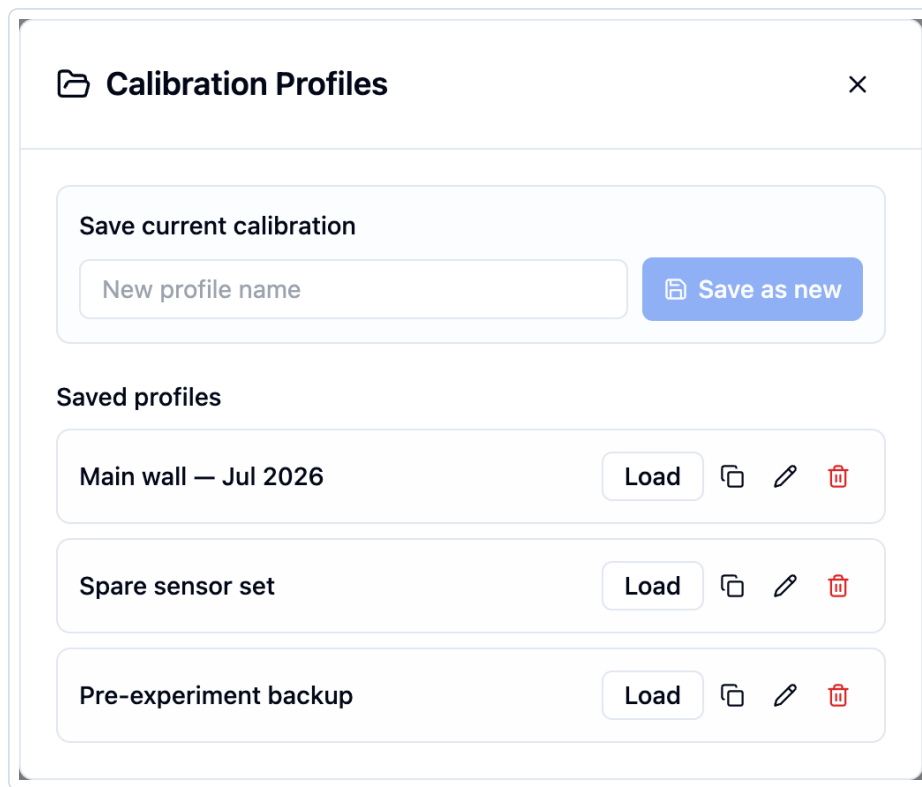


Figure 8. The Calibration Profiles panel: name and Save as new at the top, then each saved profile with Load, duplicate, rename and delete.

Saving. Type a name under Save current calibration and click Save as new to store whatever calibration is currently active. If a profile is already active, an Update “*name*” button also appears to write your latest changes back to it.

Managing. Each saved profile offers five actions:

- **Load** — apply it as the active calibration (it takes effect and is saved immediately);
- **Recalibrate** (ruler icon) — load the profile *and* open the calibration wizard on it in one click. This is the only way to change a profile’s values: they are always measured with real weights in the wizard, never typed in by hand;
- **Duplicate** — copy it under a new name;
- **Rename** — change the name only (the values stay untouched);
- **Delete** — guarded by a two-step confirmation.

The active profile is marked Active, or Active · modified once you have changed the live calibration without saving it back.

Editing a profile safely. Click the profile’s **Recalibrate** (ruler) button — it becomes the active calibration and the wizard opens on it. A banner at the top of the wizard always tells you *which* calibration you are changing (“loaded from profile ‘*name*’”) and warns once your changes differ from the saved profile. Recalibrate the boards you want — each Save updates the live calibration — then click Update “*name*” to write the changes back. Duplicate first if you want to keep the original untouched.

7. Where it is all stored

The three things from Section 1.3 are kept — or deliberately not kept — in different places:

- **The active calibration** (axis switches + scales). Pressing **Save** in the wizard writes it in three places at once: the **live program** (the next reading is already converted correctly), the browser’s **local storage** (a reload keeps it even with no backend), and a file on the backend, `calibration_settings.json` (the authoritative copy, reloaded automatically next time). Only scales and axis switches go here — **no zero**.
- **Profiles** live in `calibration_profiles.json` on the backend, so your named library survives restarts.
- **The zero (tare)** is cached *only* in the browser’s local storage. It is never persisted to the backend and never part of a calibration or profile.

Because the scales are a physical property of the sensors, you normally calibrate **once** per board; you re-zero often, but that costs nothing and saves nothing.

Do I need to recalibrate if the wall angle changes? Generally no. The scales are intrinsic to the sensors; θ only affects how a hang is split at calibration time and how forces are *displayed*. Re-running the **Hang** $\rightarrow X$ & Z step after a large angle change is only a precaution against real-world cross-axis effects. The Y axis and the overall force magnitude are unaffected either way.

8. Troubleshooting

Symptom	What to do
“No sensor data received” when capturing	Make sure the backend is running, the sensor boards are plugged in, and the dashboard is connected (Connect Device). Captures need live data and time out after 8 seconds without it.
The Zero Sensors button is greyed out	No live data is streaming. Plug in the boards, click Connect Device and wait a moment — zeroing needs about 20 samples, and demo mode cannot zero.
Unloaded readings sit at a non-zero value	Re-zero: unload the holds and click Zero Sensors. A red staleness dot means the old zero has simply drifted.
The Hang option is greyed out	The wall angle θ is set to roughly 0° . A vertical hang then puts no load on Z . Go back to Stage 2 and enter the real (non-zero) tilt.
A captured row says “swung $\pm N$ ”	The weight was still moving. Let it settle and Re-capture for a cleaner value.
Forces look wrong after calibration	First re-zero (unloaded) with Zero Sensors. If they are still off, check that θ matched the real wall and that the weights you entered match the plates you used.

Screenshots were taken from the dashboard running in demo mode; on a connected system the “No device connected” banner does not appear.